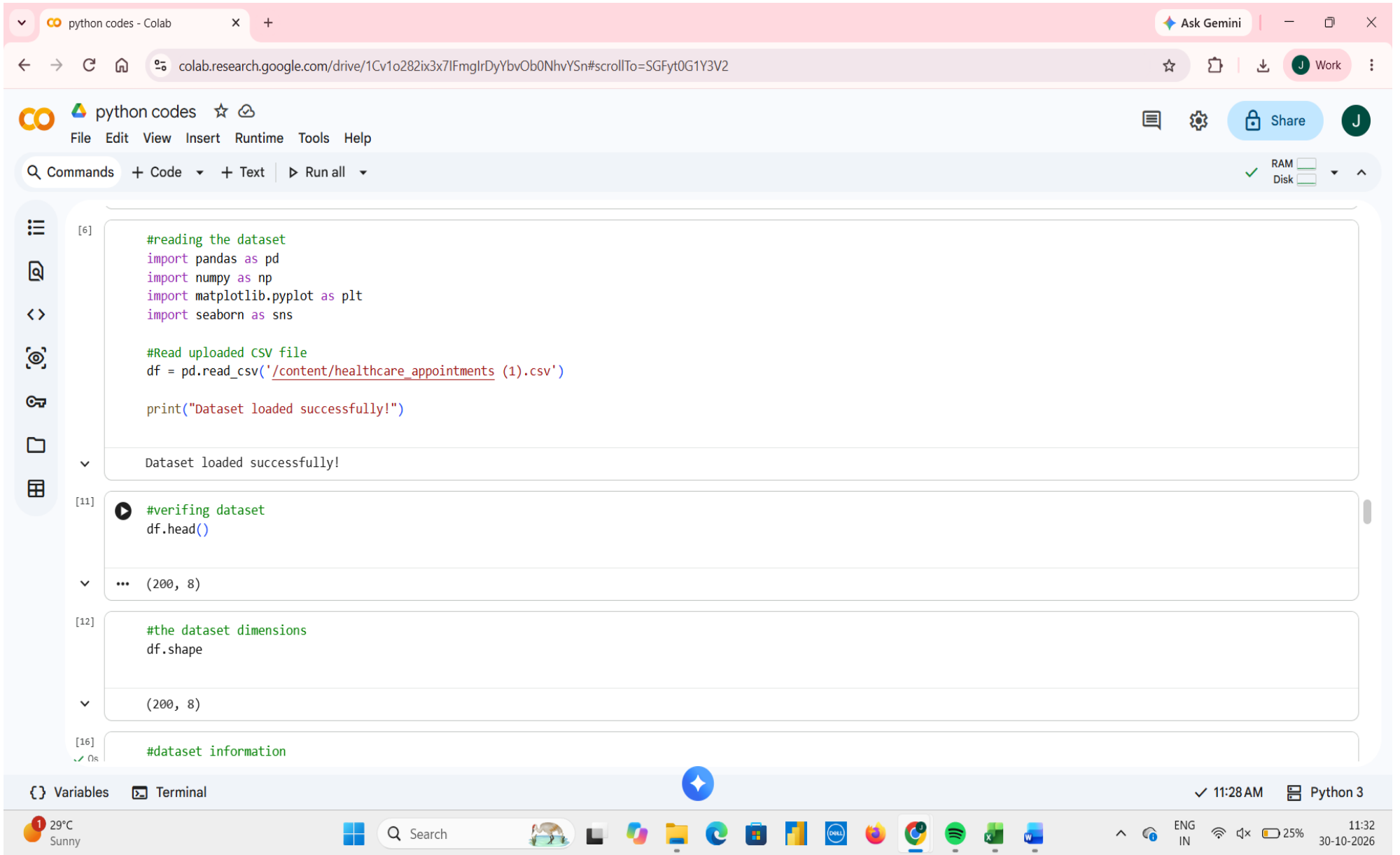
HEALTHCARE APPOINTMENTS

TRAINER - Ajey

Presented by
Akshaya
Annlea Joseph
Jeyadurga
Blesson
Rithick
Mohan Gowda
Firebird Institute of Research in Management


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[6]
#reading the dataset
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
#Read uploaded CSV file
df = pd.read_csv('/content/healthcare_appointments (1).csv')
print("Dataset loaded successfully!")
Dataset loaded successfully!
[11]
#verifing dataset
df.head()
(200, 8)
[12]
#the dataset dimensions
df.shape
(200, 8)
[16]
#dataset information
Variables
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[16] 0s

```
#dataset information
df.info()
#df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 8 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   AppointmentID     200 non-null    object
 1   Age               200 non-null    int64
 2   Gender            200 non-null    object
 3   Department        200 non-null    object
 4   VisitType         200 non-null    object
 5   Condition         200 non-null    object
 6   WaitTimeMinutes   200 non-null    int64
 7   SatisfactionScore 200 non-null    int64
dtypes: int64(3), object(5)
memory usage: 12.6+ KB
```

+ Code + Text

[15] 0s

```
#Data cleaning

df_clean = df.copy()

# 1. Correct Data Types

df_clean['AppointmentID'] = df_clean['AppointmentID'].astype('string')

numeric_columns = ['Age', 'WaitTimeMinutes', 'SatisfactionScore']
```

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```
Updated Data Types:
AppointmentID        string[python]
Age                           int64
Gender                       object
Department                   object
VisitType                    object
Condition                    object
WaitTimeMinutes               int64
SatisfactionScore             int64
dtype: object

Missing Values Before Cleaning:
AppointmentID       0
Age                 0
Gender              0
Department          0
VisitType           0
Condition           0
WaitTimeMinutes     0
SatisfactionScore   0
dtype: int64

Missing Values After Cleaning:
AppointmentID       0
Age                 0
Gender              0
Department          0
VisitType           0
Condition           0
WaitTimeMinutes     0
SatisfactionScore   0
dtype: int64
```

Variables Terminal

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```
Duplicate Records:
0

Duplicate Appointment IDs:
0

Invalid Age Records:
0

Invalid Waiting Time Records:
0

Invalid Satisfaction Scores:
0

FINAL DATA CLEANING SUMMARY
========================================
Original Records: 200
Cleaned Records: 200
Final Missing Values: 0
Final Duplicate Records: 0
Final Dataset Shape: (200, 8)

DATA CLEANING COMPLETED!
```

[18] ✓ 0s

```
# SECTION 3: STATISTICAL ANALYSIS
# HEALTHCARE APPOINTMENTS DATASET


# 1. TOTAL NUMBER OF APPOINTMENTS
```

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```
1. TOTAL APPOINTMENTS
--------------------------------------------------
Total Number of Appointments: 200

2. AVERAGE PATIENT AGE
--------------------------------------------------
Average Patient Age: 51.95 years

3. MINIMUM PATIENT AGE
--------------------------------------------------
Youngest Patient Age: 18 years

4. MAXIMUM PATIENT AGE
--------------------------------------------------
Oldest Patient Age: 85 years

5. APPOINTMENT COUNTS BY DEPARTMENT
--------------------------------------------------
Department
Cardiology          49
Neurology           40
Dermatology         38
Orthopedics         37
General Medicine    36
Name: count, dtype: int64

6. APPOINTMENT COUNTS BY GENDER
--------------------------------------------------
Gender
Other     74
Female    67
Male      59
```

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```
7. APPOINTMENT COUNTS BY VISIT TYPE
--------------------------------------------------
VisitType
Routine      77
New          63
Follow-up    60
Name: count, dtype: int64

8. APPOINTMENT COUNTS BY PATIENT CONDITION
--------------------------------------------------
Condition
Migraine            31
Asthma              30
Seasonal Allergy    30
Back Pain           26
Diabetes            23
Hypertension        23
Routine Checkup     22
Arthritis           15
Name: count, dtype: int64

9. AVERAGE WAITING TIME
--------------------------------------------------
Average Waiting Time: 58.07 minutes

10. AVERAGE PATIENT SATISFACTION
--------------------------------------------------
Average Satisfaction Score: 3.01 out of 5

11. MEDIAN PATIENT AGE
--------------------------------------------------
Median Patient Age: 52.0
```

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```
12. MEDIAN WAITING TIME
--------------------------------------------------
Median Waiting Time: 55.0 minutes

13. DEPARTMENT-WISE APPOINTMENT PERCENTAGE
--------------------------------------------------
Department
Cardiology          24.5
Neurology           20.0
Dermatology         19.0
Orthopedics         18.5
General Medicine    18.0
Name: proportion, dtype: float64

14. FINAL STATISTICAL SUMMARY
==================================================
       Statistical Measure  Value
        Total Appointments 200.00
       Average Patient Age  51.95
       Minimum Patient Age  18.00
       Maximum Patient Age  85.00
        Median Patient Age  52.00
      Average Waiting Time  58.07
       Median Waiting Time  55.00
Average Satisfaction Score   3.01

STATISTICAL ANALYSIS COMPLETED SUCCESSFULLY!
```

[19] ✓ 0s

```
# ==================================================
# SECTION 4: DATA MANIPULATION
```

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```
1. DATA FILTERING
================================================

Patients Above 50 Years:
  AppointmentID  Age  Gender   Department  VisitType         Condition  \
1       APT0002   72   Other  Dermatology  Follow-up   Routine Checkup
2       APT0003   75  Female  Orthopedics    Routine  Seasonal Allergy
5       APT0006   59    Male   Cardiology  Follow-up  Seasonal Allergy
7       APT0008   68  Female  Orthopedics        New  Seasonal Allergy
8       APT0009   80  Female    Neurology  Follow-up  Seasonal Allergy

   WaitTimeMinutes  SatisfactionScore
1               53                  2
2               92                  3
5               45                  3
7               16                  5
8              101                  5

Patients Waiting More Than 60 Minutes:
   AppointmentID  Age  Gender   Department  VisitType         Condition  \
0       APT0001   22  Female  Dermatology    Routine            Asthma
2       APT0003   75  Female  Orthopedics    Routine  Seasonal Allergy
4       APT0005   24  Female  Orthopedics    Routine            Asthma
8       APT0009   80  Female    Neurology  Follow-up  Seasonal Allergy
10      APT0011   33    Male  Orthopedics    Routine   Routine Checkup

    WaitTimeMinutes  SatisfactionScore
0               118                  1
2                92                  3
4               106                  2
8               101                  5
10               98                  1
```

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```
8             101                3
10             98                1

Patients With Satisfaction Above 3:
   AppointmentID  Age  Gender       Department  VisitType        Condition  \
7        APT0008   68  Female      Orthopedics        New  Seasonal Allergy
8        APT0009   80  Female       Neurology  Follow-up  Seasonal Allergy
9        APT0010   69  Female      Orthopedics  Follow-up           Asthma
12       APT0013   64  Female      Orthopedics    Routine   Routine Checkup
21       APT0022   42   Other  General Medicine  Follow-up           Asthma

    WaitTimeMinutes  SatisfactionScore
7                16                  5
8               101                  5
9                35                  4
12               60                  4
21               29                  4

2. ASCENDING SORTING
================================================

Patients Sorted by Age - Ascending:
    AppointmentID  Age  Gender       Department  VisitType        Condition  \
41        APT0042   18    Male  General Medicine  Follow-up   Routine Checkup
159       APT0160   18    Male      Orthopedics    Routine   Routine Checkup
106       APT0107   19    Male       Cardiology  Follow-up  Seasonal Allergy
6         APT0007   19   Other      Dermatology        New           Asthma
136       APT0137   19   Other       Neurology    Routine  Seasonal Allergy
93        APT0094   19   Other      Orthopedics        New     Hypertension
61        APT0062   20   Other       Neurology        New         Migraine
111       APT0112   20   Other  General Medicine    Routine   Routine Checkup
166       APT0167   20  Female       Cardiology    Routine        Back Pain
```

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```
149       APT0150   20  Female       Dermatology        New      Hypertension

     WaitTimeMinutes  SatisfactionScore
41                65                  2
159               37                  1
106               98                  5
6                 31                  3
136               66                  5
93                74                  4
61               110                  2
111               50                  3
166               48                  4
149                9                  3

Waiting Time - Ascending:
    AppointmentID  Age  Gender        Department  VisitType         Condition  \
147       APT0148   32    Male       Orthopedics    Routine  Seasonal Allergy
143       APT0144   72    Male       Dermatology        New          Diabetes
80        APT0081   84  Female        Cardiology        New          Diabetes
144       APT0145   65  Female  General Medicine    Routine         Back Pain
181       APT0182   22   Other         Neurology  Follow-up      Hypertension
102       APT0103   28   Other       Dermatology        New          Migraine
89        APT0090   35   Other       Orthopedics        New   Routine Checkup
73        APT0074   77  Female        Cardiology    Routine  Seasonal Allergy
138       APT0139   52  Female  General Medicine  Follow-up  Seasonal Allergy
28        APT0029   23    Male         Neurology        New         Arthritis

     WaitTimeMinutes  SatisfactionScore
147                0                  3
143                0                  4
80                 2                  3
144                2                  2
```

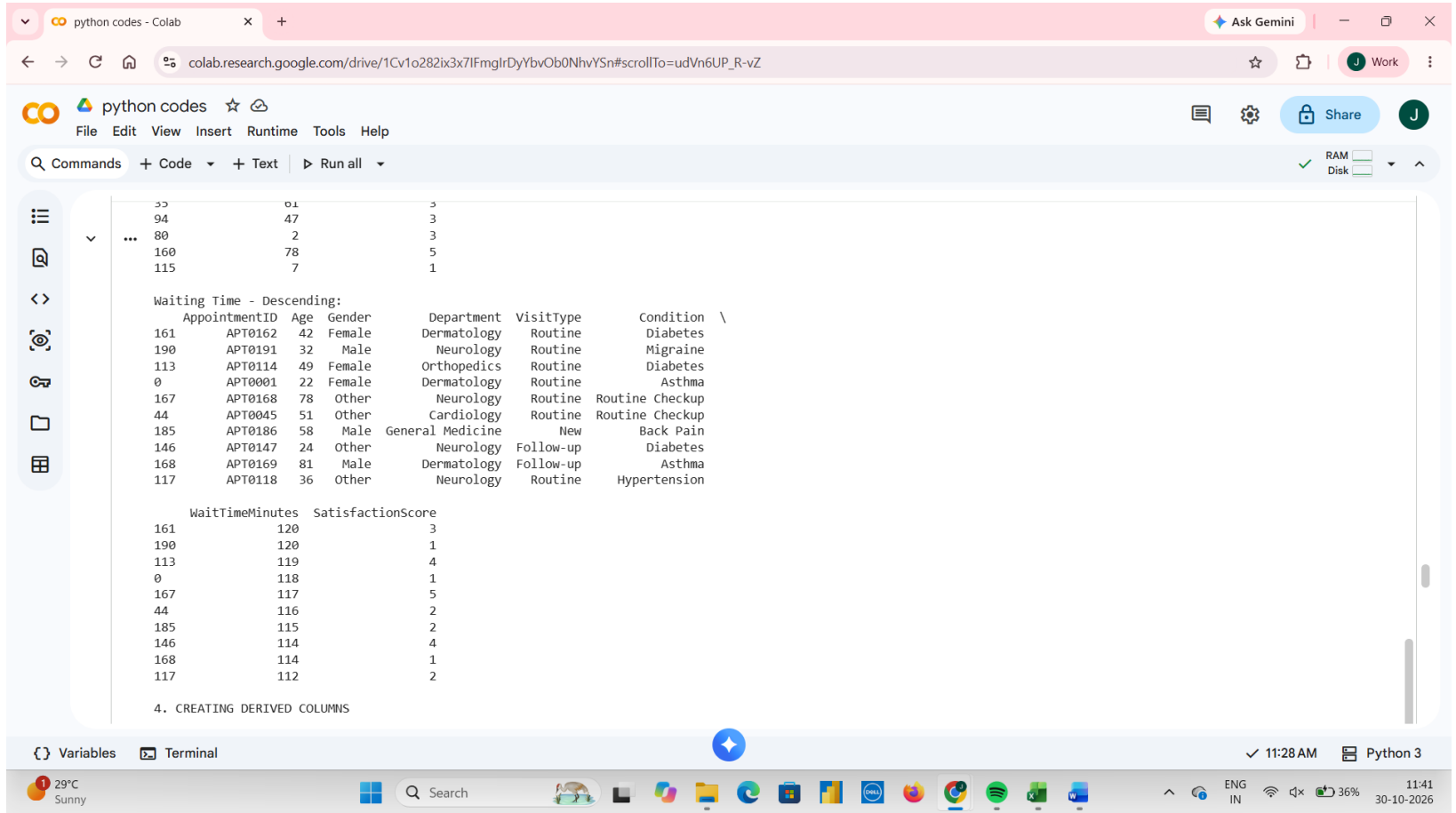
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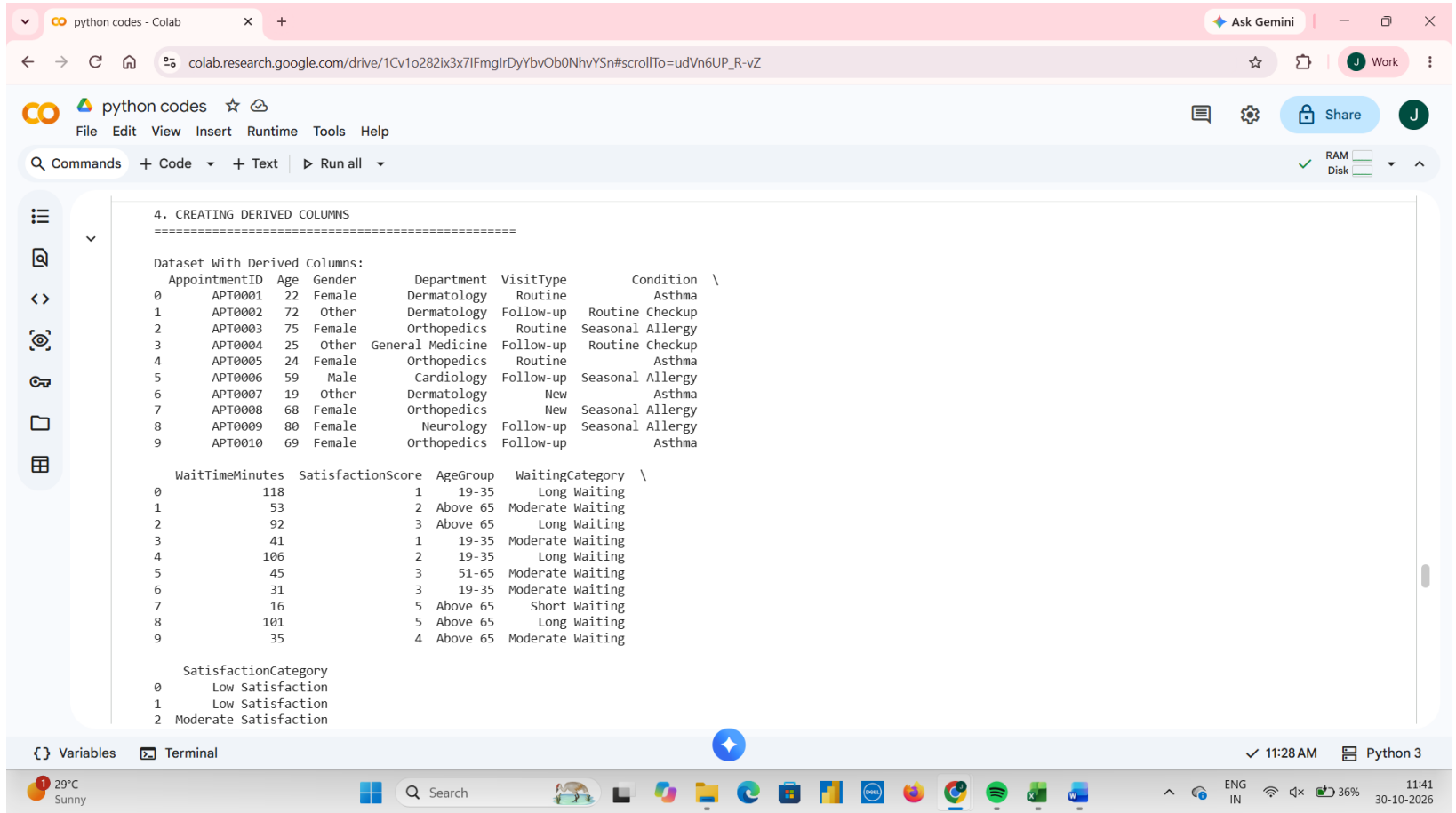
```
144                2                 2
181                2                 2
102                3                 1
89                 4                 2
73                 5                 1
138                5                 2
28                 7                 1

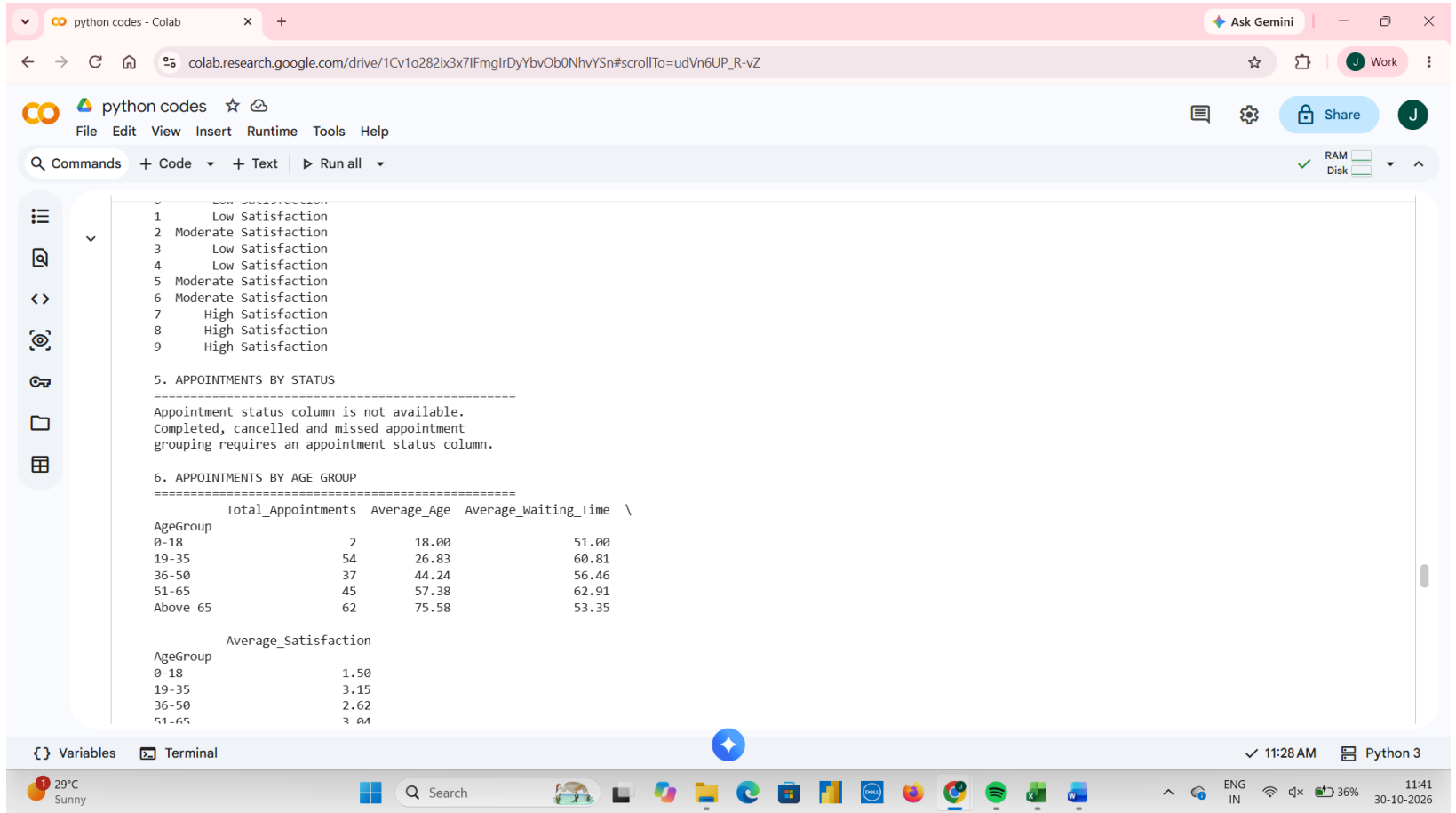
3. DESCENDING SORTING
==================================================

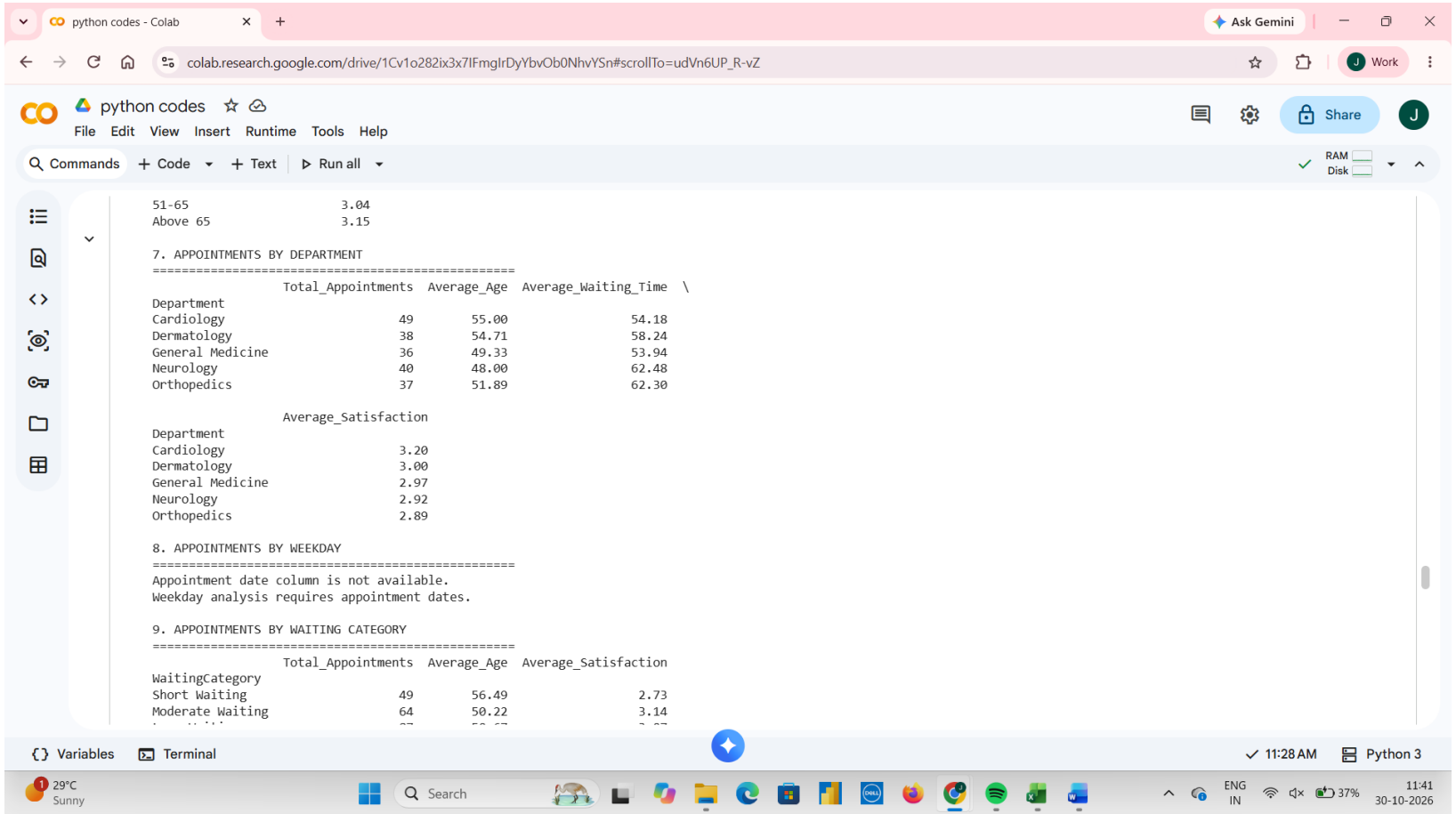
Patients Sorted by Age - Descending:
    AppointmentID  Age  Gender        Department  VisitType        Condition  \
139       APT0140   85  Female         Neurology        New     Hypertension
194       APT0195   85    Male        Cardiology    Routine     Hypertension
172       APT0173   85    Male  General Medicine  Follow-up           Asthma
175       APT0176   85   Other        Cardiology  Follow-up        Back Pain
59        APT0060   84  Female        Cardiology    Routine           Asthma
35        APT0036   84   Other  General Medicine    Routine         Migraine
94        APT0095   84   Other        Cardiology        New        Back Pain
80        APT0081   84  Female        Cardiology        New         Diabetes
160       APT0161   84   Other         Neurology  Follow-up  Routine Checkup
115       APT0116   83    Male  General Medicine        New        Arthritis

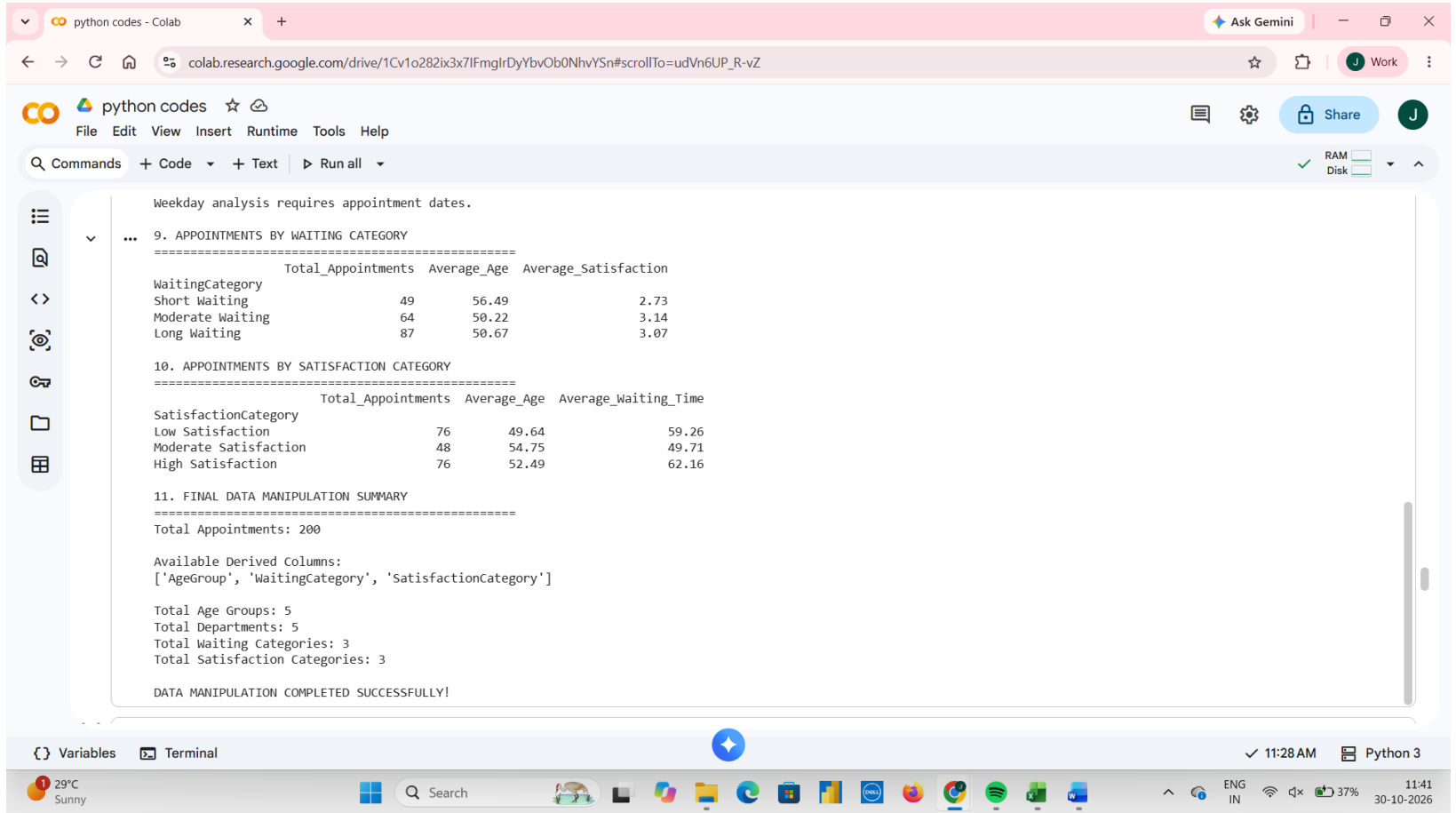
     WaitTimeMinutes  SatisfactionScore
139               77                  3
194               82                  4
172               86                  1
175               16                  2
59                38                  3
35                61                  3
94                47                  3
```


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RAM
Disk
35 61 3
94 47 3
80 2 3
160 78 5
115 7 1
Waiting Time - Descending:
AppointmentID Age Gender Department VisitType Condition \
161 APT0162 42 Female Dermatology Routine Diabetes
190 APT0191 32 Male Neurology Routine Migraine
113 APT0114 49 Female Orthopedics Routine Diabetes
0 APT0001 22 Female Dermatology Routine Asthma
167 APT0168 78 Other Neurology Routine Routine Checkup
44 APT0045 51 Other Cardiology Routine Routine Checkup
185 APT0186 58 Male General Medicine New Back Pain
146 APT0147 24 Other Neurology Follow-up Diabetes
168 APT0169 81 Male Dermatology Follow-up Asthma
117 APT0118 36 Other Neurology Routine Hypertension
WaitTimeMinutes SatisfactionScore
161 120 3
190 120 1
113 119 4
0 118 1
167 117 5
44 116 2
185 115 2
146 114 4
168 114 1
117 112 2
4. CREATING DERIVED COLUMNS
Variables
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4. CREATING DERIVED COLUMNS
==
Dataset With Derived Columns:
AppointmentID Age Gender Department VisitType Condition \
0 APT0001 22 Female Dermatology Routine Asthma
1 APT0002 72 Other Dermatology Follow-up Routine Checkup
2 APT0003 75 Female Orthopedics Routine Seasonal Allergy
3 APT0004 25 Other General Medicine Follow-up Routine Checkup
4 APT0005 24 Female Orthopedics Routine Asthma
5 APT0006 59 Male Cardiology Follow-up Seasonal Allergy
6 APT0007 19 Other Dermatology New Asthma
7 APT0008 68 Female Orthopedics New Seasonal Allergy
8 APT0009 80 Female Neurology Follow-up Seasonal Allergy
9 APT0010 69 Female Orthopedics Follow-up Asthma
WaitTimeMinutes SatisfactionScore AgeGroup WaitingCategory \
0 118 1 19-35 Long Waiting
1 53 2 Above 65 Moderate Waiting
2 92 3 Above 65 Long Waiting
3 41 1 19-35 Moderate Waiting
4 106 2 19-35 Long Waiting
5 45 3 51-65 Moderate Waiting
6 31 3 19-35 Moderate Waiting
7 16 5 Above 65 Short Waiting
8 101 5 Above 65 Long Waiting
9 35 4 Above 65 Moderate Waiting
SatisfactionCategory
0 Low Satisfaction
1 Low Satisfaction
2 Moderate Satisfaction
Variables
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RAM
Disk
1 Low Satisfaction
2 Moderate Satisfaction
3 Low Satisfaction
4 Low Satisfaction
5 Moderate Satisfaction
6 Moderate Satisfaction
7 High Satisfaction
8 High Satisfaction
9 High Satisfaction
5. APPOINTMENTS BY STATUS
Appointment status column is not available.
Completed, cancelled and missed appointment
grouping requires an appointment status column.
6. APPOINTMENTS BY AGE GROUP
Total_Appointments Average_Age Average_Waiting_Time \
AgeGroup
0-18 2 18.00 51.00
19-35 54 26.83 60.81
36-50 37 44.24 56.46
51-65 45 57.38 62.91
Above 65 62 75.58 53.35
Average_Satisfaction
AgeGroup
0-18 1.50
19-35 3.15
36-50 2.62
Variables
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Work
python codes
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Commands + Code + Text Run all
RAM
Disk
51-65 3.04
Above 65 3.15
7. APPOINTMENTS BY DEPARTMENT
==
Total_Appointments Average_Age Average_Waiting_Time \
Department
Cardiology 49 55.00 54.18
Dermatology 38 54.71 58.24
General Medicine 36 49.33 53.94
Neurology 40 48.00 62.48
Orthopedics 37 51.89 62.30
Average_Satisfaction
Department
Cardiology 3.20
Dermatology 3.00
General Medicine 2.97
Neurology 2.92
Orthopedics 2.89
8. APPOINTMENTS BY WEEKDAY
==
Appointment date column is not available.
Weekday analysis requires appointment dates.
9. APPOINTMENTS BY WAITING CATEGORY
==
Total_Appointments Average_Age Average_Satisfaction
WaitingCategory
Short Waiting 49 56.49 2.73
Moderate Waiting 64 50.22 3.14
Variables
Terminal
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RAM
Disk
Weekday analysis requires appointment dates.
9. APPOINTMENTS BY WAITING CATEGORY
==
Total_Appointments Average_Age Average_Satisfaction
WaitingCategory
Short Waiting 49 56.49 2.73
Moderate Waiting 64 50.22 3.14
Long Waiting 87 50.67 3.07
10. APPOINTMENTS BY SATISFACTION CATEGORY
==
Total_Appointments Average_Age Average_Waiting_Time
SatisfactionCategory
Low Satisfaction 76 49.64 59.26
Moderate Satisfaction 48 54.75 49.71
High Satisfaction 76 52.49 62.16
11. FINAL DATA MANIPULATION SUMMARY
==
Total Appointments: 200
Available Derived Columns:
['AgeGroup', 'WaitingCategory', 'SatisfactionCategory']
Total Age Groups: 5
Total Departments: 5
Total Waiting Categories: 3
Total Satisfaction Categories: 3
DATA MANIPULATION COMPLETED SUCCESSFULLY!
Variables
Terminal
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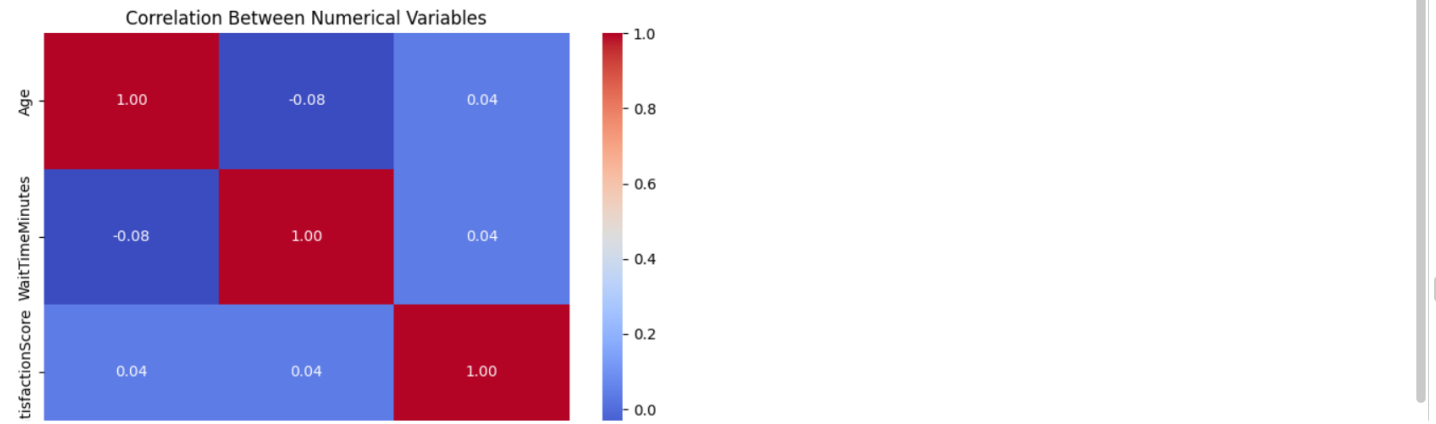
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RAM Disk

```
Cannot analyze appointment trends.
The dataset does not contain an appointment date column.

2. CORRELATION ANALYSIS
Correlation Matrix:
                        Age  WaitTimeMinutes  SatisfactionScore
Age                1.000000        -0.081989           0.037905
WaitTimeMinutes   -0.081989         1.000000           0.037680
SatisfactionScore  0.037905         0.037680           1.000000
```



Variables Terminal

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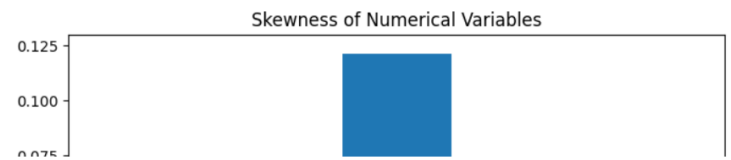
```
3. OUTLIER DETECTION

Outliers in Age:
Lower Limit: -19.125
Upper Limit: 121.875
Number of Outliers: 0

Outliers in WaitTimeMinutes:
Lower Limit: -52.625
Upper Limit: 170.375
Number of Outliers: 0

Outliers in SatisfactionScore:
Lower Limit: -1.0
Upper Limit: 7.0
Number of Outliers: 0

4. SKEWNESS ANALYSIS
Skewness of Numerical Columns:
Age                -0.046406
WaitTimeMinutes     0.121387
SatisfactionScore   0.007044
dtype: float64
```



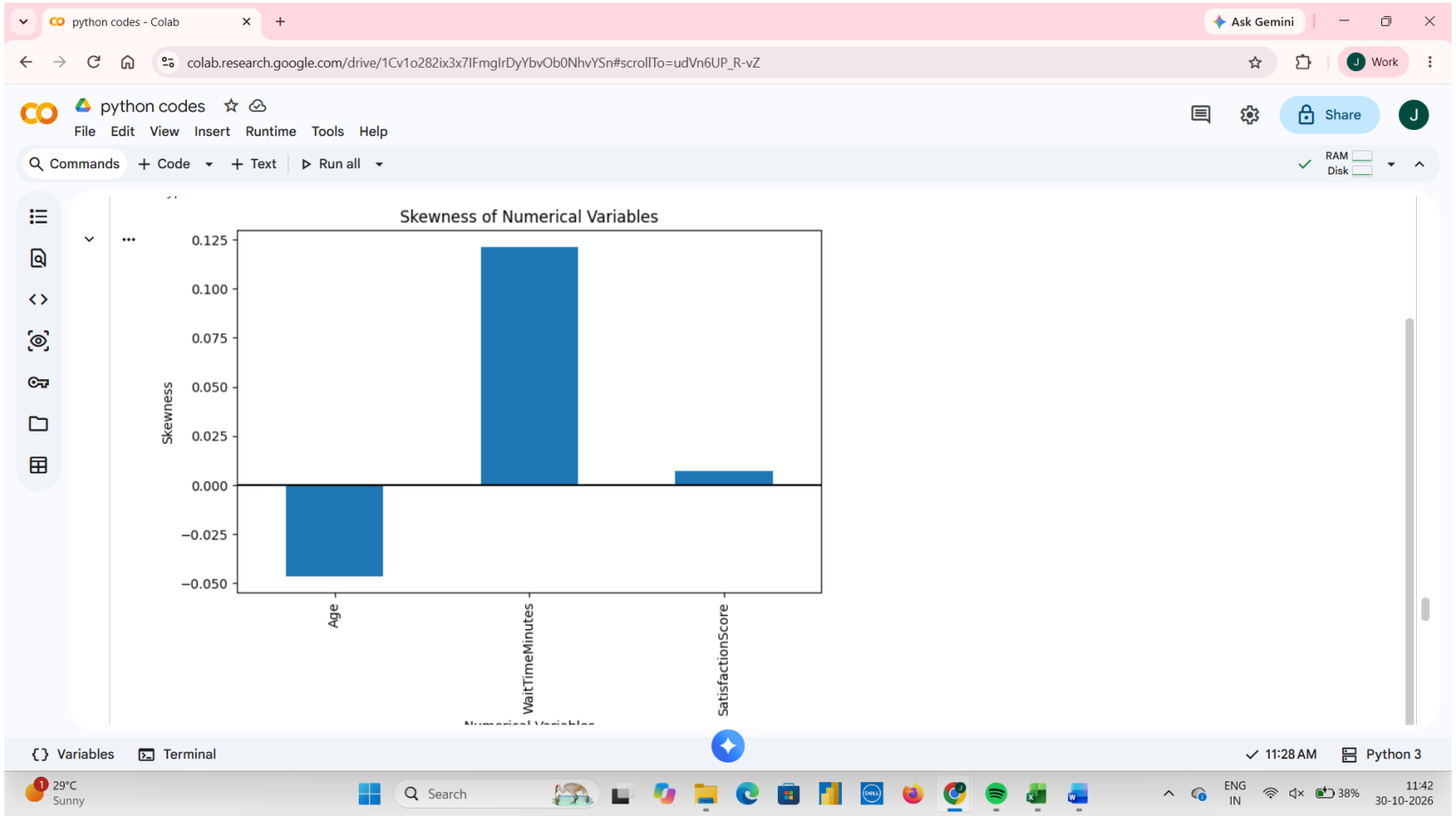
Variables Terminal

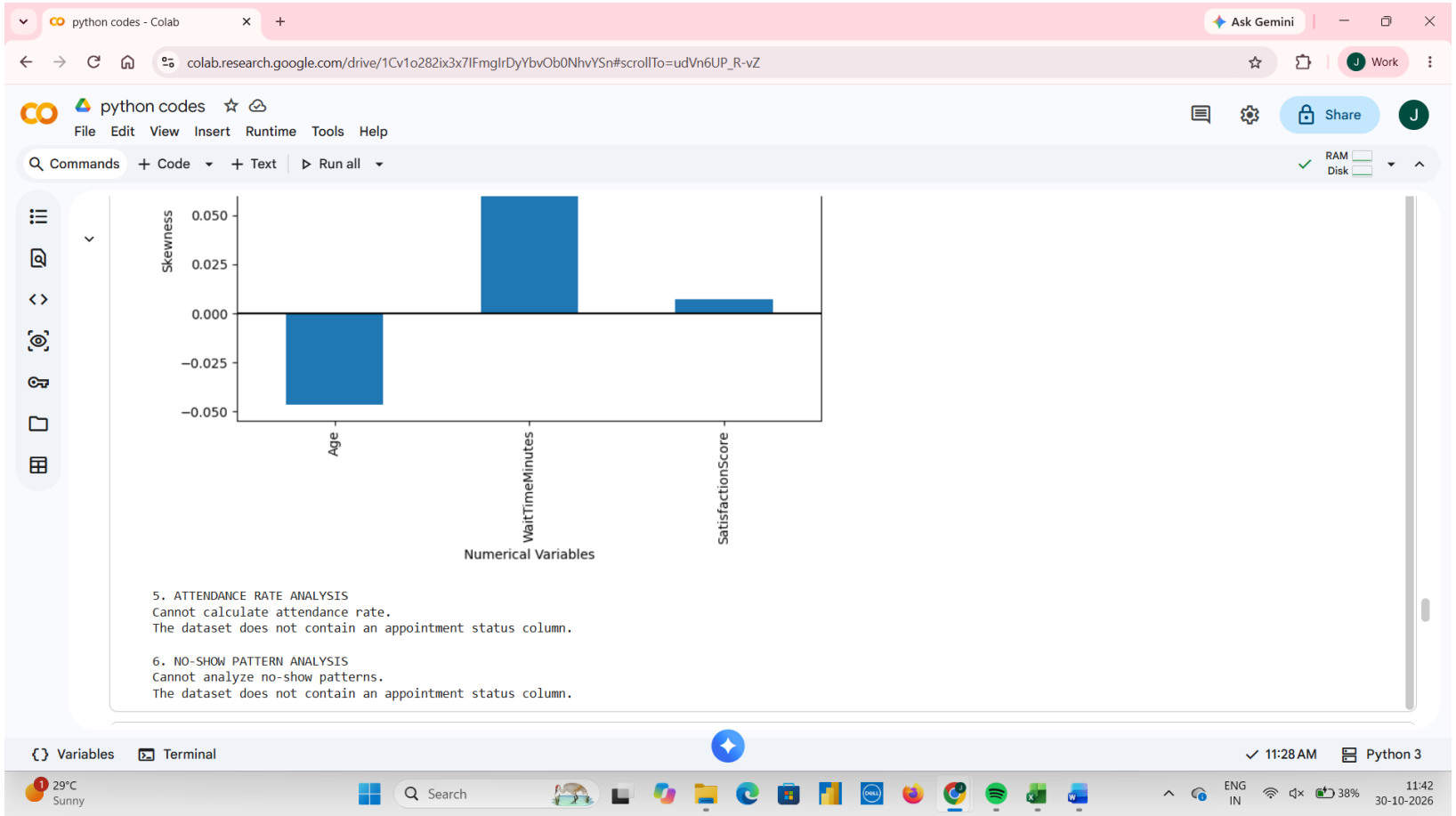
11:28 AM Python 3

29°C Sunny

Search

ENG IN 38% 11:42 30-10-2026


python codes - Colab
Ask Gemini
colab.research.google.com/drive/1Cv1o282ix3x7IFmgIrDyYbvOb0NhvYSn#scrollTo=udVn6UP_R-vZ
Work
python codes
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM
Disk
Skewness of Numerical Variables
Skewness
0.125
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
Age
WaitTimeMinutes
SatisfactionScore
Variables
Terminal
11:28 AM
Python 3
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Work
python codes
File Edit View Insert Runtime Tools Help
Share
Commands + Code + Text Run all
RAM
Disk
Skewness
0.050
0.025
0.000
−0.025
−0.050
Age
WaitTimeMinutes
SatisfactionScore
Numerical Variables
5. ATTENDANCE RATE ANALYSIS
Cannot calculate attendance rate.
The dataset does not contain an appointment status column.
6. NO-SHOW PATTERN ANALYSIS
Cannot analyze no-show patterns.
The dataset does not contain an appointment status column.
Variables
Terminal
11:28 AM
Python 3
29°C
Sunny
Search
ENG
IN
38%
11:42
30-10-2026

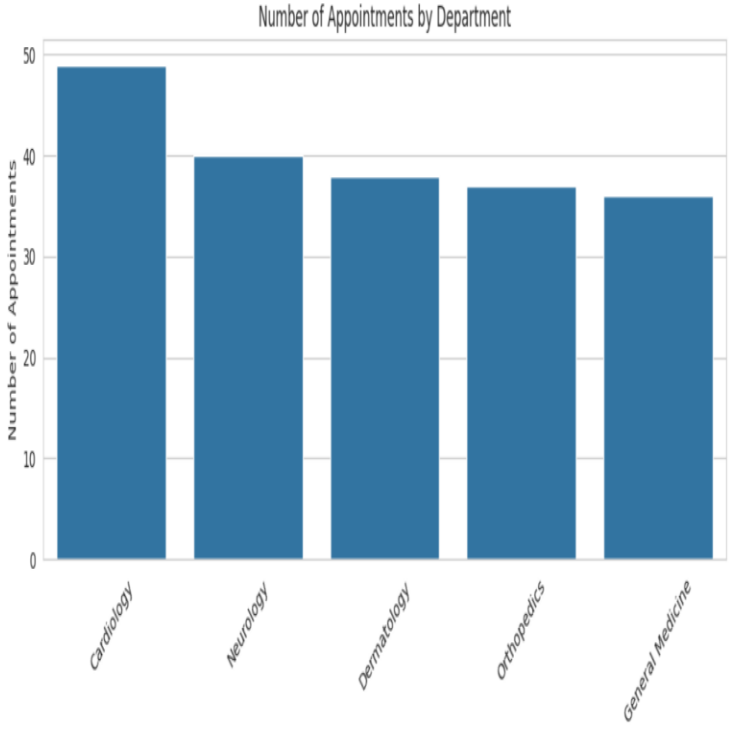
python codes - Colab

colab.research.google.com/drive/1Cv1o282ix3x7IFmgIrDyYbvOb0NhvYSn#scrollTo=udVn6UP_R-vZ

python codes

File Edit View Insert Runtime Tools Help

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Variables Terminal

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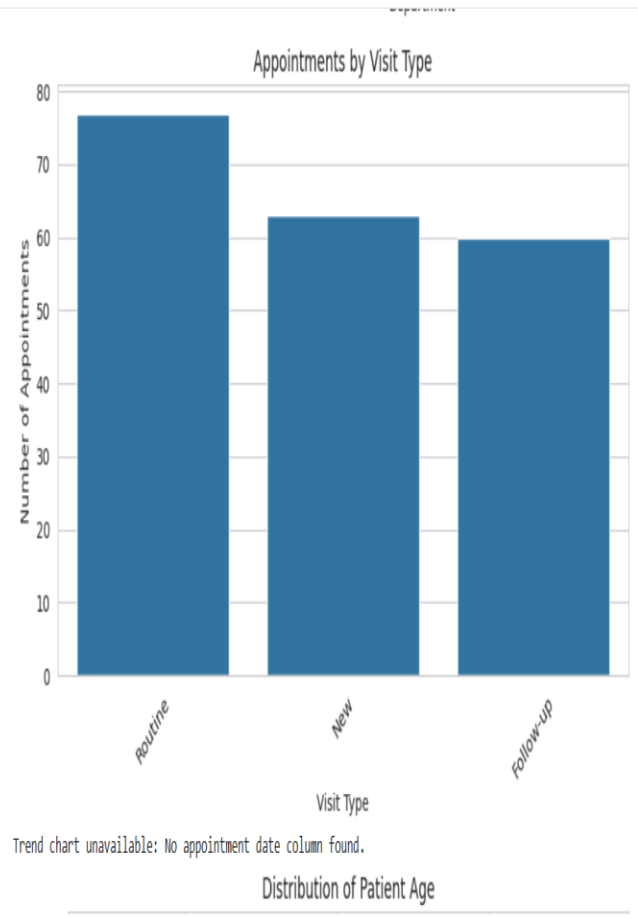
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39%

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Trend chart unavailable: No appointment date column found.

Distribution of Patient Age

python codes - Colab

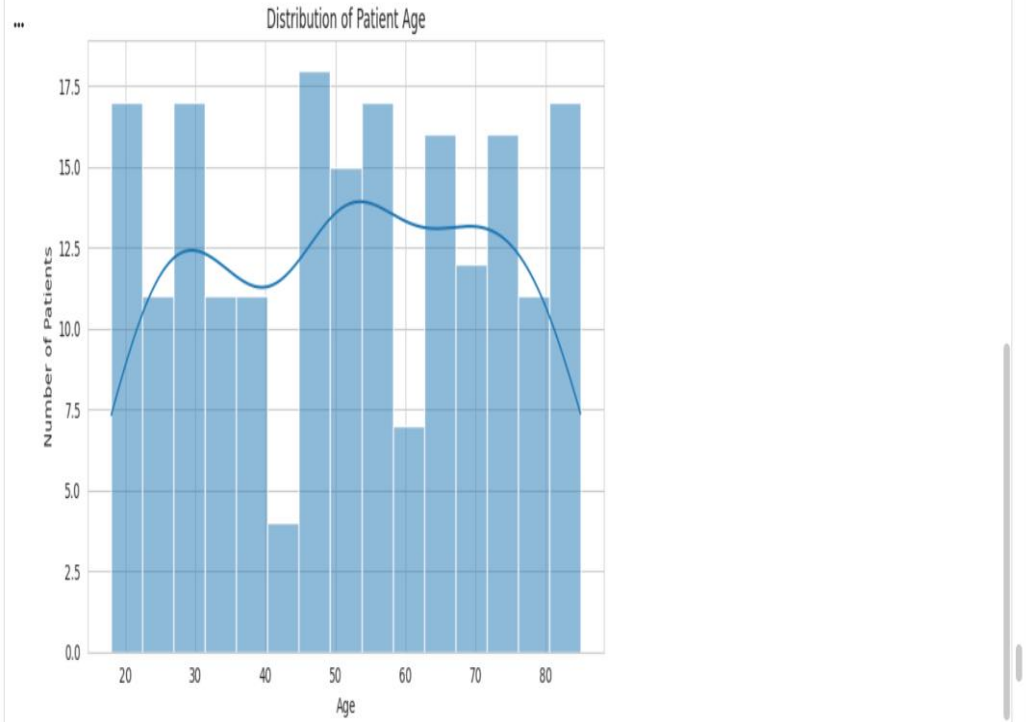
colab.research.google.com/drive/1Cv1o282ix3x7IFmgIrDyYbvOb0NhvYSn#scrollTo=udVn6UP_R-vZ

python codes

File Edit View Insert Runtime Tools Help

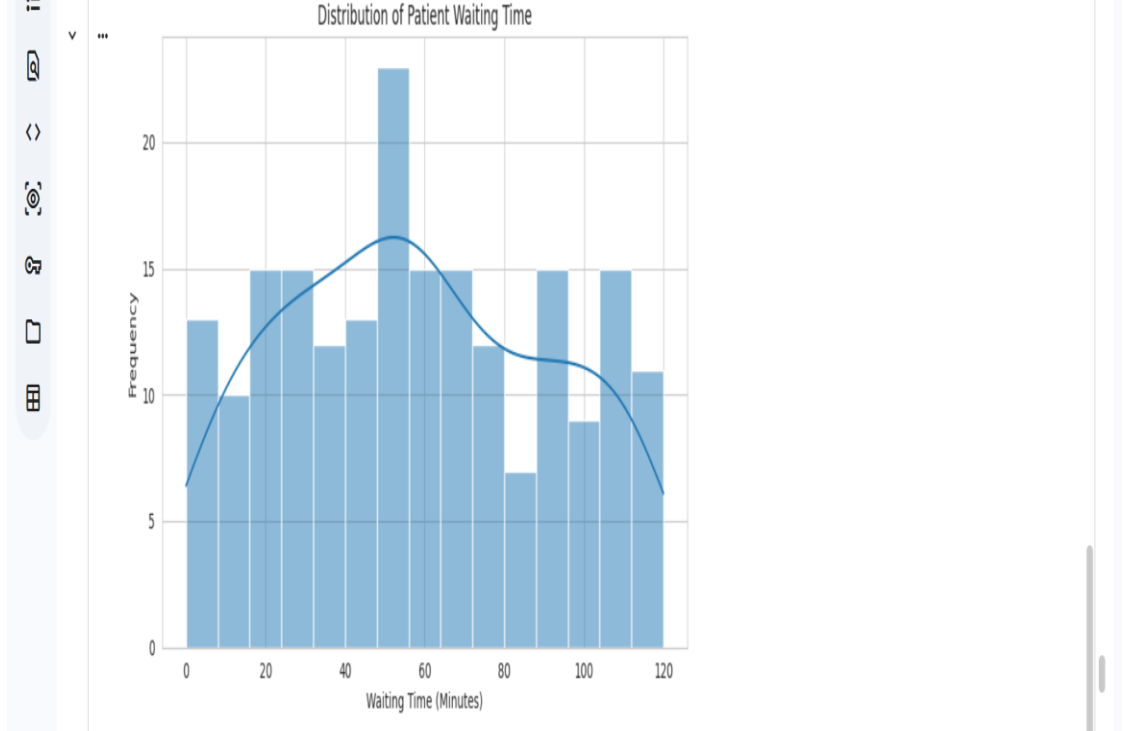
Commands + Code + Text Run all

Trend chart unavailable: No appointment date column found.



Distribution of Patient Waiting Time

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Relationship Between Age and Waiting Time

120

Variables Terminal 11:28 AM Python 3

python codes - Colab

colab.research.google.com/drive/1Cv1o282ix3x7IFmgIrDyYbvOb0NhvYSn#scrollTo=udVn6UP_R-vZ

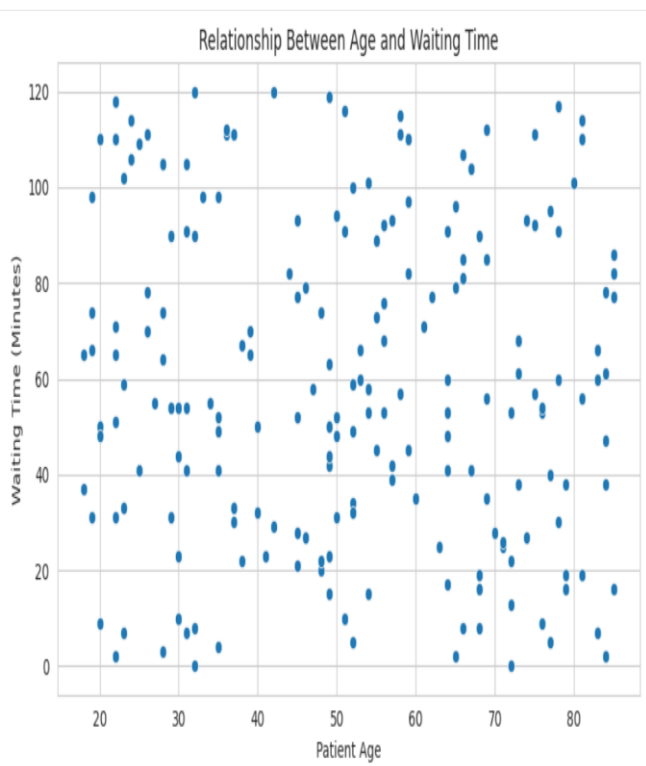
python codes

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colab.research.google.com/drive/1Cv1o282ix3x7IFmgIrDyYbvOb0NhvYSn#scrollTo=udVn6UP_R-vZ

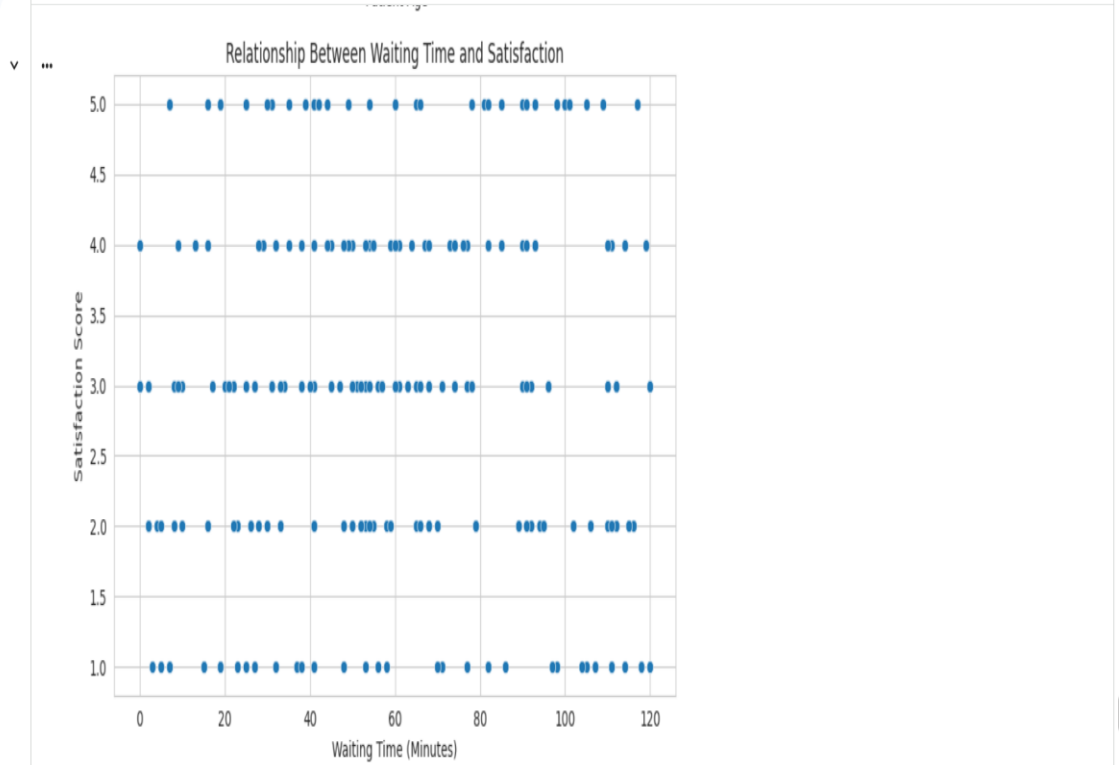
python codes

File Edit View Insert Runtime Tools Help

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RAM Disk

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Variables Terminal

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Search

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python codes - Colab

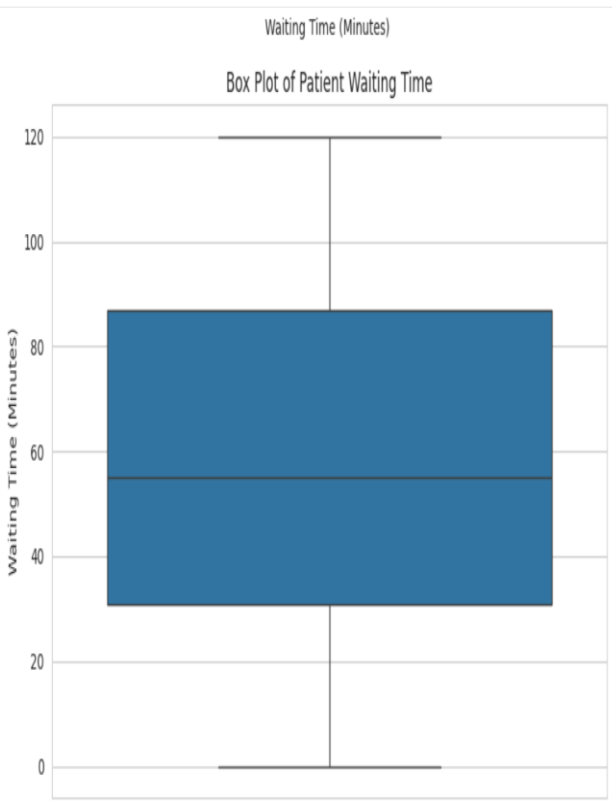
colab.research.google.com/drive/1Cv1o282ix3x7IFmgIrDyYbvOb0NhvYSn#scrollTo=udVn6UP_R-vZ

python codes

File Edit View Insert Runtime Tools Help

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Waiting Time (Minutes)

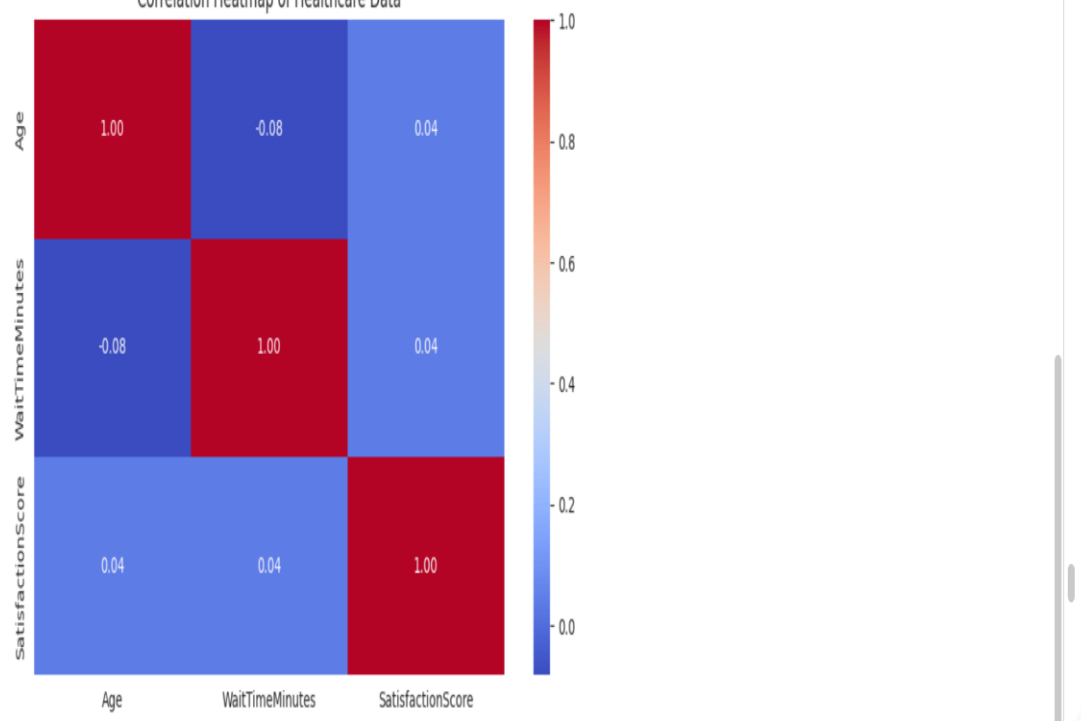


Correlation Heatmap of Healthcare Data

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python codes - Colab

colab.research.google.com/drive/1Cv1o282ix3x7IFmgIrDyYbvOb0NhvYSn#scrollTo=udVn6UP_R-vZ

python codes

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
1. DEPARTMENT-WISE APPOINTMENTS
                Total Appointments
Department
Cardiology                      49
Dermatology                     38
General Medicine                36
Neurology                       40
Orthopedics                     37

2. APPOINTMENTS BY VISIT TYPE
           Total Appointments
VisitType
Follow-up                  60
New                        63
Routine                    77

3. APPOINTMENTS BY GENDER
         Total Appointments
Gender
Female                   67
Male                     59
Other                    74

4. DEPARTMENT VS VISIT TYPE
VisitType        Follow-up  New  Routine  Grand Total
Department
Cardiology              13   18       18           49
Dermatology             11   12       15           38
General Medicine        12   13       11           36
Neurology               17    8       15           40
Orthopedics              7   12       18           37
```

Variables Terminal 11:28 AM Python 3

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python codes - Colab

colab.research.google.com/drive/1Cv1o282ix3x7IFmgIrDyYbvOb0NhvYSn#scrollTo=udVn6UP_R-vZ

python codes

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
Grand Total            60   63       77          200

5. DEPARTMENT PERFORMANCE
                      Age  SatisfactionScore  WaitTimeMinutes
Department
Cardiology      55.000000           3.204082        54.183673
Dermatology     54.710526           3.000000        58.236842
General Medicine 49.333333          2.972222        53.944444
Neurology       48.000000           2.925000        62.475000
Orthopedics     51.891892           2.891892        62.297297

6. MERGING DEPARTMENT DATASET
No separate department dataset found.
Upload department_data.csv to perform this merge.

7. MERGING DOCTOR DATASET
No separate doctor dataset found.
Upload doctor_data.csv to perform this merge.

8. VALIDATING APPOINTMENT TOTALS
Total Appointments in Dataset: 200
Total from Department Pivot: 200
Total from Visit Type Pivot: 200
Total from Gender Pivot: 200

9. VALIDATION RESULTS
Department Total Matches: True
Visit Type Total Matches: True
Gender Total Matches: True

10. DUPLICATE APPOINTMENT ID CHECK
Duplicate Appointment IDs: 0
```

Variables Terminal 11:28 AM Python 3

29°C Sunny Search ENG IN 41% 11:43 30-10-2026

python codes - Colab

colab.research.google.com/drive/1Cv1o282ix3x7lFmgIrDyYbvOb0NhvYSn#scrollTo=udVn6UP_R-vZ

python codes

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
10. DUPLICATE APPOINTMENT ID CHECK
Duplicate Appointment IDs: 0
All appointment IDs are unique.

11. MISSING VALUE VALIDATION
AppointmentID        0
Age                  0
Gender               0
Department           0
VisitType            0
Condition            0
WaitTimeMinutes      0
SatisfactionScore    0
AgeGroup             0
WaitingCategory      0
SatisfactionCategory 0
dtype: int64
No missing values found.

12. FINAL VALIDATION SUMMARY
                 Validation  Result
0       Total Appointments     200
1    Department Pivot Total    200
2    Visit Type Pivot Total    200
3       Gender Pivot Total     200
4  Duplicate Appointment IDs     0
5      Total Missing Values      0
```

[23] ✓ 0s

```
# ========================================
```

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python codes - Colab

python codes

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Commands + Code + Text Run all

colab.research.google.com/drive/1Cv1o282ix3x7lFmgIrDyYbvOb0NhvYSn#scrollTo=SGFyt0G1Y3V2

python codes

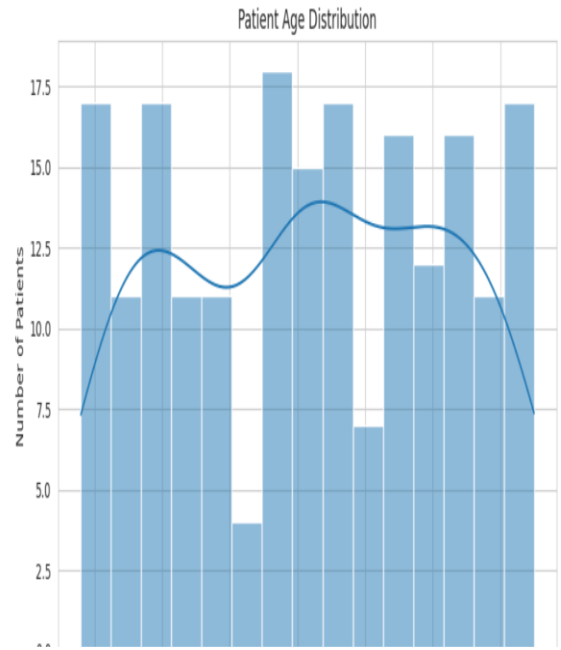
File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
1. BAR CHART - APPOINTMENT STATUS
Bar chart unavailable: Appointment status column not found.

2. LINE CHART - MONTHLY APPOINTMENT TRENDS
Line chart unavailable: Appointment date column not found.

3. HISTOGRAM - PATIENT AGE DISTRIBUTION
```



Variables Terminal ✓ 11:28 AM Python 3

29°C Sunny Search ENG IN 42% 11:44 30-10-2026

python codes - Colab

colab.research.google.com/drive/1Cv1o282ix3x7lFmgIrDyYbvOb0NhvYSn#scrollTo=SGFyt0G1Y3V2

python codes

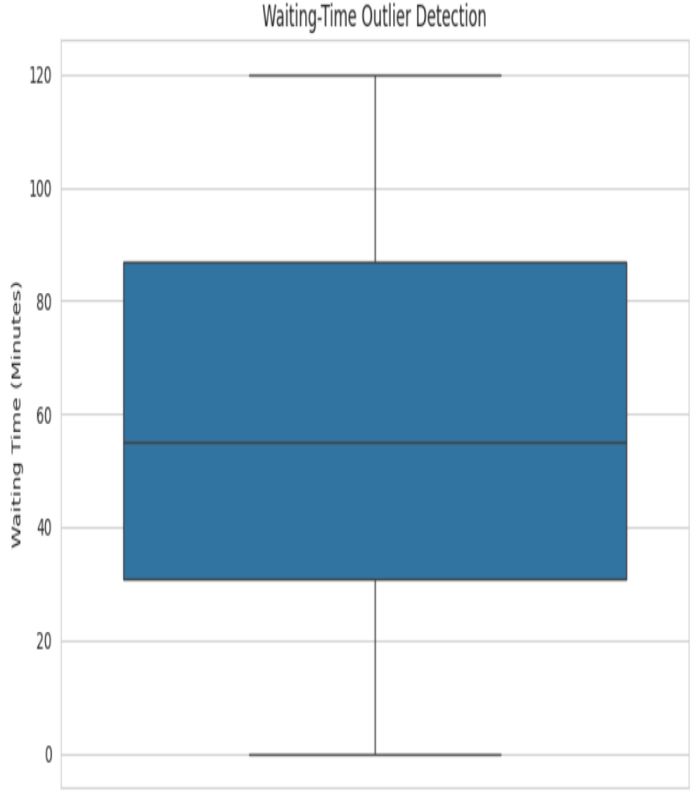
File Edit View Insert Runtime Tools Help

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5. BOX PLOT - WAITING TIME OUTLIERS



6. HEATMAP - CORRELATION ANALYSIS

Variables Terminal

11:28 AM Python 3

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python codes - Colab

colab.research.google.com/drive/1Cv1o282ix3x7lFmgIrDyYbvOb0NhvYSn#scrollTo=SGFyt0G1Y3V2

python codes

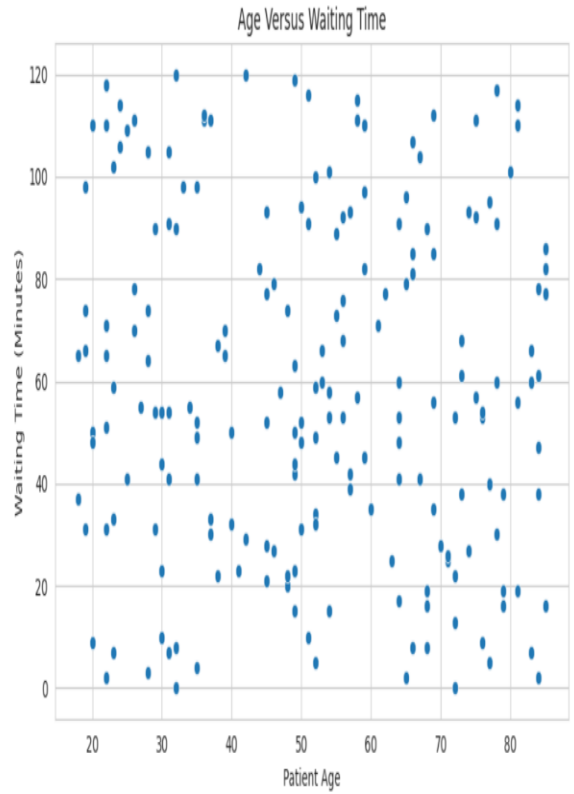
File Edit View Insert Runtime Tools Help

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RAM Disk

4. SCATTER PLOT - AGE VS WAITING TIME



5. BOX PLOT - WAITING TIME OUTLIERS

Variables Terminal

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colab.research.google.com/drive/1Cv1o282ix3x7IFmgIrDyYbvOb0NhvYSn#scrollTo=SGFyt0G1Y3V2

Work

python codes

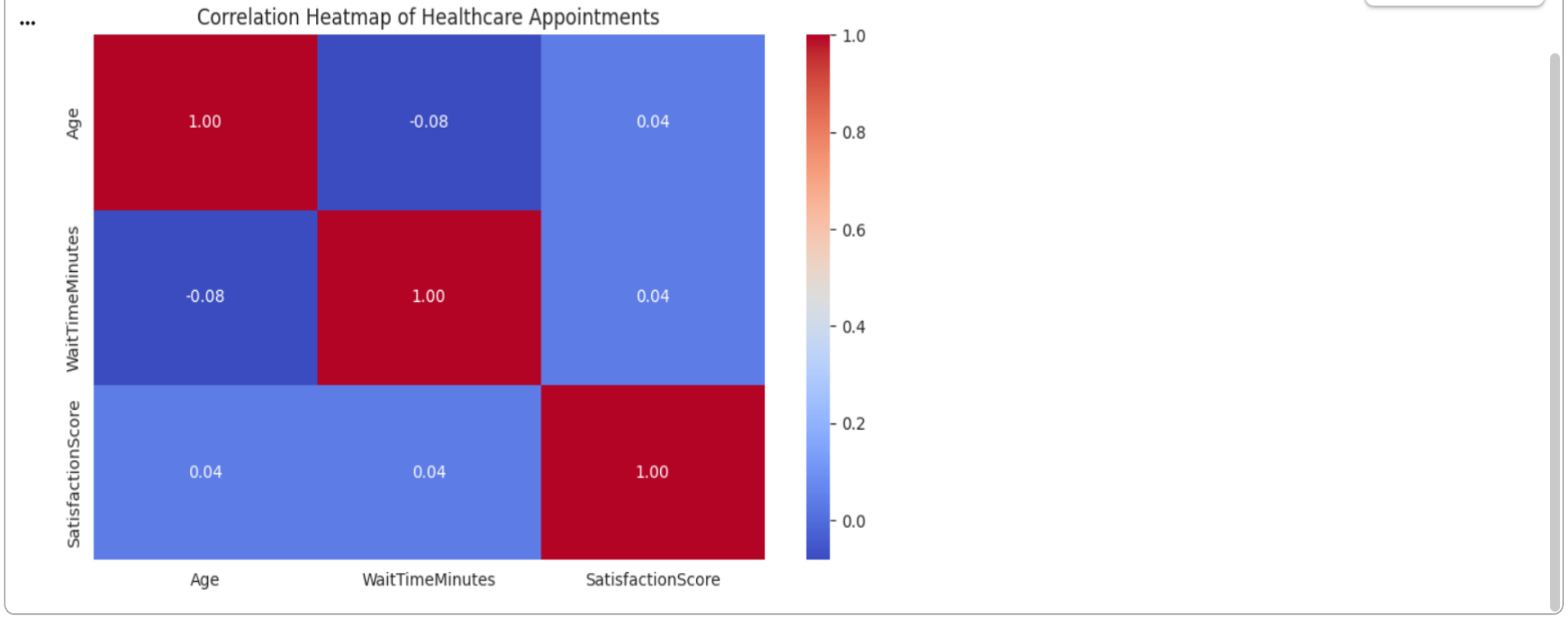
File Edit View Insert Runtime Tools Help

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RAM Disk

6. HEATMAP - CORRELATION ANALYSIS



Variables Terminal

11:28 AM Python 3

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ENG IN

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python codes - Colab

Ask Gemini

colab.research.google.com/drive/1Cv1o282ix3x7IFmgIrDyYbvOb0NhvYSn#scrollTo=UTAkX6id2OIJ

Work

python codes

File Edit View Insert Runtime Tools Help

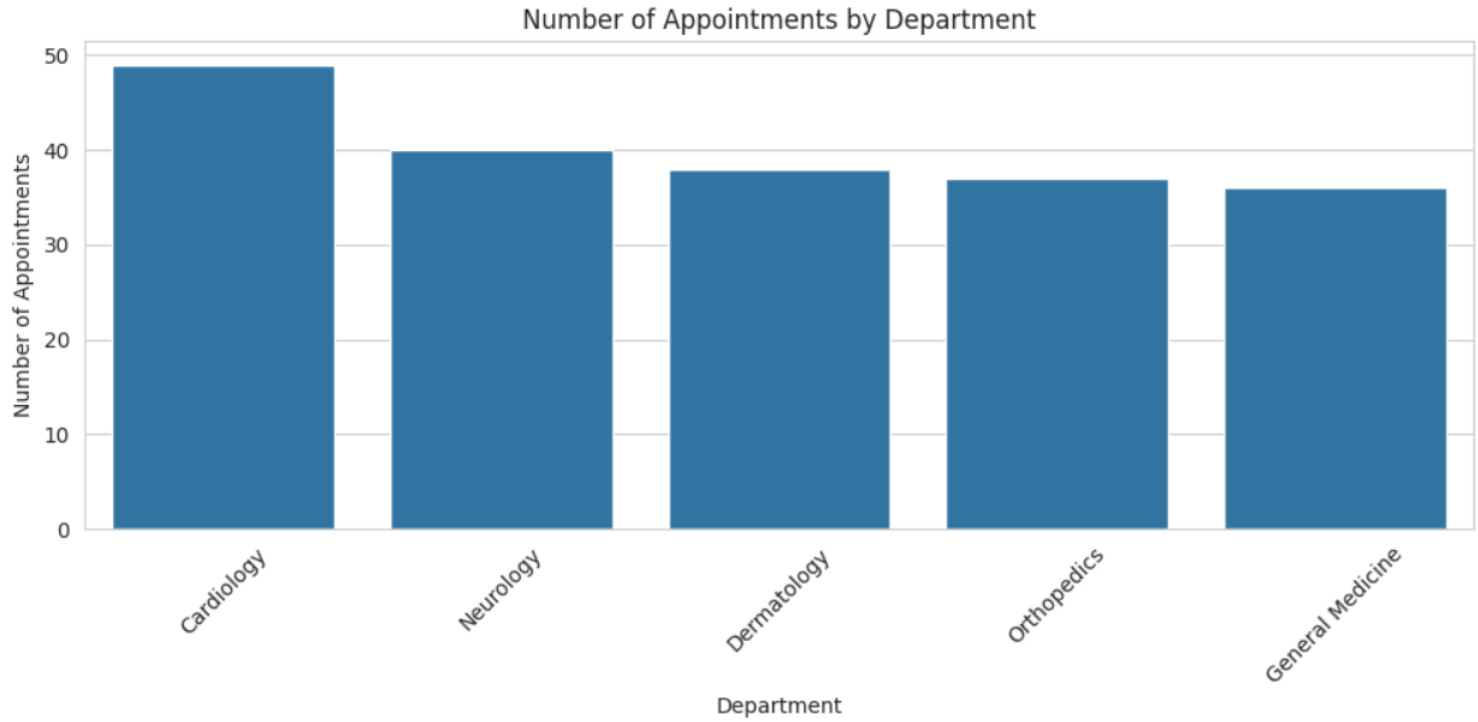
Share

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RAM Disk

[24]

```
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Variables Terminal

1:37 PM Python 3

33°C Sunny

Search

ENG IN

93%

13:40 30-10-2026

python codes - Colab

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Work

python codes

File Edit View Insert Runtime Tools Help

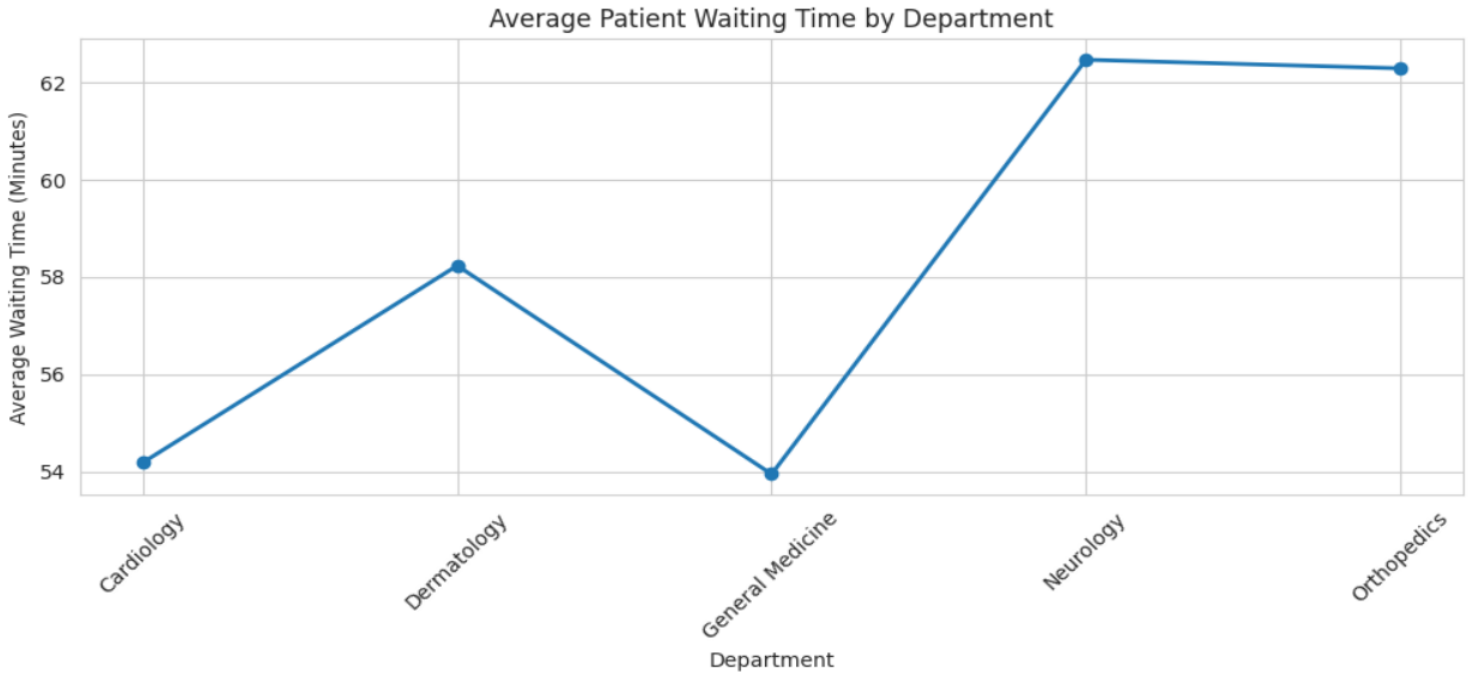
Share

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RAM Disk

[25]

```
plt.show()
```



Variables Terminal

1:37 PM Python 3

33°C Sunny

Search

ENG IN 92% 13:40 30-10-2026

THANK YOUU FOR YOUR PATIENT LISTENING